1 A preorder on condensations from coding-recondensation

How should we judge which condensation of the random variables $(X_i)_{i\in I}$ is best? We can imagine each of two different condensations having some specific merits over the other, each perhaps making explicit some structure latent in $(X_i)_{i\in I}$ which the other one does not properly reveal. But maybe one condensation being better than another could still be well-captured by some preorder on entropy tuples? Unfortunately for the existence of a fully satisfactory such preorder, it seems plausible that there could even be some specific condensation $(Y_A)_{A\in\mathcal{P}^+I}$ which is more complex than another condensation $(Z_A)_{A\in\mathcal{P}^+I}$ at each A — i.e., for which for all $A\in\mathcal{P}^+I$, we have $H(Y_A)\geq H(Z_A)$ (or $H(Y_A|(Y_B)_{B\supsetneq A})\geq H(Z_A|(Z_B)_{B\supsetneq A})$ if we are working with conditional scores) — but which still reveals something that $(Z_A)_{A\in\mathcal{P}^+I}$ does not. Still, we would like to have some judgment, so we will accept sometimes calling a condensation worse than another despite it having some merits. Here is a proposal for a preorder on condensations $(Y_A)_{A\in\mathcal{P}^+I}$ induced by a preorder $\geq$ on their entropy vectors $(H(Y_A))_{A\in\mathcal{P}^+I}\in\mathbb{R}^{\mathcal{P}^+I}$:[1]

Preliminary definition 1.1. We could say that $\mathbf{h}\geq\mathbf{g}$ for $\mathbf{h},\mathbf{g}\in\mathbb{R}^{\mathcal{P}^+I}$ if and only if whenever some random variables $(X_i)_{i\in I}$ have a condensation at entropies $\mathbf{h}$, they also have a condensation at entropies $\mathbf{g}$.

** explanation why this isn't the right definition and one should ask the coding question instead. idk if i should explain the coding setup here **

Definition 1.1 (The recondensation poset)**.** We say that $\mathbf{h}\geq_C\mathbf{g}$ for $\mathbf{h},\mathbf{g}\in\mathbb{R}^{\mathcal{P}^+I}$ if and only if whenever some random variables $(X_i)_{i\in I}$ have a condensation at entropies $\mathbf{h}$, they can be coded at rates $\mathbf{g}$.

actually the thing below should probably be replaced by just proving it is given by a cone. the main claim here is that if

Proposition 1. *The relation $\geq_C$ is a partial order on $\mathbb{R}_{\geq 0}^{\mathcal{P}^+I}$. That is, $\geq_C$ is reflexive, antisymmetric, and transitive.*

Proof. We prove that $\geq_C$ is reflexive, antisymmetric, and transitive:

$\geq_C$ is reflexive: This is just saying that variables with entropy $\mathbf{h}$ can be coded at rates $\mathbf{h}$, which follows from Shannon's source coding theorem.

$\geq_C$ is antisymmetric: We want to show that if $\mathbf{h}\geq\mathbf{g}$ and $\mathbf{g}\geq\mathbf{h}$, then $\mathbf{h}=\mathbf{g}$. This follows by getting inequalities in each direction on subset-upset sums from perfect condensations and noting they together imply pointwise equality.

$\geq_C$ is transitive: If $\mathbf{h}\geq_C\mathbf{g}$ and $\mathbf{g}\geq_C\mathbf{f}$, then one can code any random variables with entropies $\mathbf{h}$ at rates $\mathbf{g}$. todo: actually here it seems like i should establish that the relation interacts well with the algebraic structure first

[1] to do: the story works for unconditioned entropies but i haven't thought carefully about whether it also works for conditioned entropies — i should clarify this later.

□

The ordering of the various condensations of $(X_i)_{i \in I}$ induced by comparing their entropy vectors will not be antisymmetric, though, as there are meaningfully different condensations with the same entropies. But it will of course be reflexive and transitive, so it is a preorder.

The following theorem gives some alternative characterizations of the poset $\geq_C$ on $\mathbb{R}_{\geq 0}^{\mathcal{P}^+ I}$.

Theorem 1. *The following are equivalent:*

$\mathbf{h} \geq_C \mathbf{g}$ *(definition):* *Whenever some random variables $(X_i)_{i \in I}$ have a condensation at entropies $\mathbf{h}$, they can be coded at rates $\mathbf{g}$. That is, $\mathbf{h} \geq_C \mathbf{g}$.*

Coding to coding: *Whenever some random variables $(X_i)_{i \in I}$ can be coded at rates $\mathbf{h}$, they can be coded at rates $\mathbf{g}$.*

- *def just for a single perfect condensation ground variable tuple*
- *some kind of copying thing?*
- *given by pos cone*

** explaining that $\geq_C$ is given by a positive cone **

Definition 1.2. The recondensation cone

1.1 The coding partial order is bigger than the up-set partial order and smaller than the subset-up-set partial order

1.2 The recondensation problem is equivalent to a hypergraph coding problem

what should we mean by reductions even? maybe i should just write up a sketch of the argument and then decide?

but like: I guess we care about whether one can code at certain capacities with arbitrarily small error probability. or maybe we should say arbitrarily close to these capacities at arbitrarily small error probability? depends on whether we want an open or closed set i guess. either way, probably the most interesting equivalence

the reduction: the condensation coding problem is obviously a special case of the hypergraph coding problem. but the hypergraph coding problem can be reduced to condensation coding also, as follows. the ground variables will be in bijection with the edges of the hypergraph — that is, the index set I will just be the set E of edges of the hypergraph. the ground variables will be perfect condensation variables given by certain independent latents. these latents will correspond to the source variables of the hypergraph coding problem, i.e. the colors of its edge coloring, and the index set at which the latent for a color lives is the set of all edges of that color. the new variables correspond to vertices of the hypergraph, ie to channels of the hypergraph coding problem. the index set for the variable for a vertex is the set of all edges adjacent to that vertex. now the requirements of the coding-condensation problem

is there some issue with 0 edges here? like can the rate region change because we have these other channels with rate 0 now? i guess we could say you have to stick to the rates precisely and achieve negligible error probability? would that break anything? hmm since i think we're in the collocated source case of network coding here, from `https://youtu.be/K0lPYFRrp44?t=1473` it seems plausible this won't actually matter for us — i think we should probably get the same rate region if we allow deviations by δ from the capacities, ie only ask for negligible error in the presence of any small deviation up in capacities. ahh yep you can just send everywhere what you'd want to send in the missing edge everywhere. you can compute it because sources are collocated, so just compute it at the start

1.3 Solutions to the hypergraph coding problem for some small graphs

1.4 The hypergraph coding problem is equivalent to the network coding problem

Proposition 2. *Hypergraph coding reduces to network coding.*

Proof. the reduction gadget has a single source seeing everything, then □

Proposition 3. *Hypergraph coding reduces to network coding.*

Proof. thing with a vertex for each channel □

need to look at the defn of the index coding problem in `https://arxiv.org/pdf/1211.6660` and then pick a defn for our coding problem to make everything hang together

This also provides some reason to think that the problem of telling which entropies one can perform condensation at is hard.